S A Aravind Eswar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.49
Sri Chaitanya Techno School	Class XII (CBSE)	2024	95.40%
Sri Chaitanya Techno School	Class X (CBSE)	2022	92.20%

RELEVANT COURSES

Electrical Engineering: Digital Systems (Theory and Lab), DSP Lab, Semiconductor Device Fundamentals, Introduction to CMOS Processing, Analog Electronics

Mathematics: Probability, Random Processes, Machine Learning, Signal Processing, Numerical Analysis, Differential Equations and Transform Techniques, Vector Calculus, Introduction to Statistics

SKILLS

Languages:

Tools and Technology:

PROJECTS

Asymmetric Asynchronous FIFO - GitHub

- **Engineered an Asynchronous Asymmetric FIFO** to facilitate robust data transfer across independent clock domains, implementing AXI stream protocol signals (TVALID, TREADY, TLAST,TKEEP) for seamless peripheral communication.
- **Synthesized and profiled the architecture using Intel Quartus**, leveraging design methodologies from SNUG industry papers to optimize resource consumption and resolve strict timing constraints, achieving max read/write clock frequencies exceeding 200MHz+.

DIF FFT Module - GitHub

- **Designed an 8-point Decimation-in-Frequency FFT** using cascaded Single-path Delay Feedback (SDF) submodules with serial I/O, and validated frequency-domain outputs against theoretical DFT results across multiple input frequencies on an Altera MAX 10 FPGA.
- **Achieved 99.36 MHz Fmax using only 735 logic elements, outperforming Quartus's built-in FFT IP** core (69.77 MHz, 2,219 logic elements) in both speed and resource utilization by **over 3x.**

Complex Eigenvalue Computer - GitHub

- **Implemented the QR algorithm from scratch in C to compute eigenvalues of complex-valued matrices**, using Givens rotations for QR decomposition and accelerating convergence via Hessenberg reduction and Wilkinson shifts.
- **Parallelized matrix operations with OpenMP and benchmarked three optimization strategies** (naive, Hessenberg-only, shifted Hessenberg), identifying shifted Hessenberg as fastest; used Valgrind to identify and fix memory leaks in dynamic matrix allocation.

Raycast Renderer - GitHub

- **Built a real-time raycasting engine in C++/SDL2** implementing DDA-based wall casting and separate floor/ceiling texture mapping, with custom BMP texture loading and a frame-buffer-based rendering pipeline.
- Parallelized per-pixel ray computations using OpenMP across wall, floor, and ceiling render passes, achieving real-time frame rates by distributing independent ray-cast workloads across CPU threads.

Scientific Calculator - GitHub

- **Engineered an ATmega328P-based scientific calculator**, developing bare-metal firmware in Embedded C (via AVR-GCC/AVRDUDE) and hand-soldering a custom 10x3 button matrix onto a protoboard for the physical interface.
- **Implemented the hardware-efficient CORDIC algorithm** to execute complex mathematical operations, including trigonometric, hyperbolic, and exponential functions, optimizing microcontroller processing speed and resource

utilization.

Alarm Clock (RTL) - GitHub

- **Co-developed a comprehensive Verilog alarm clock and timer system** within a three-person team, engineering adaptable hardware datapaths to maintain accurate timekeeping across both 12-hour and 24-hour display formats.
- **Architected the core control logic utilizing a Finite State Machine (FSM)** to efficiently manage complex state transitions, processing three dedicated user inputs for seamless mode switching, alarm configuration, and time adjustment.

POSITION OF RESPONSIBILITY

Senior Core - Robotix

Events and Logistics Coordinator - Vibes

EXPERIENCE

Research Intern | IITH Oct 2025-May 2026

- **Conducted algorithmic research under Dr. Anjana at IITH** to conceptualize an automated music remixing pipeline, focusing on quantifying song similarity through probabilistic tagging.
- **Engineered and benchmarked diverse audio feature extraction techniques**, comparatively analyzing Short-Time Fourier Transform (STFT), Mel scales, and MFCCs to evaluate signal processing efficacy and data richness.
- **Evaluated lightweight machine learning classifiers** (including KNN, GMMs, and SVMs) to optimize computational efficiency, deriving actionable architectural insights for generating and assigning probabilistic similarity scores to discrete audio tracks.

EXTRACURRICULAR ACTIVITIES

Guitarist - Performed at various club events and fests within college as part of a student band.